

ABSTRACT ID: IG18-A001

Generative AI with Kernel-adaptive Diffusion for High-resolution Satellite Image Inpainting Across Urban, Forest, and Agricultural Landscapes

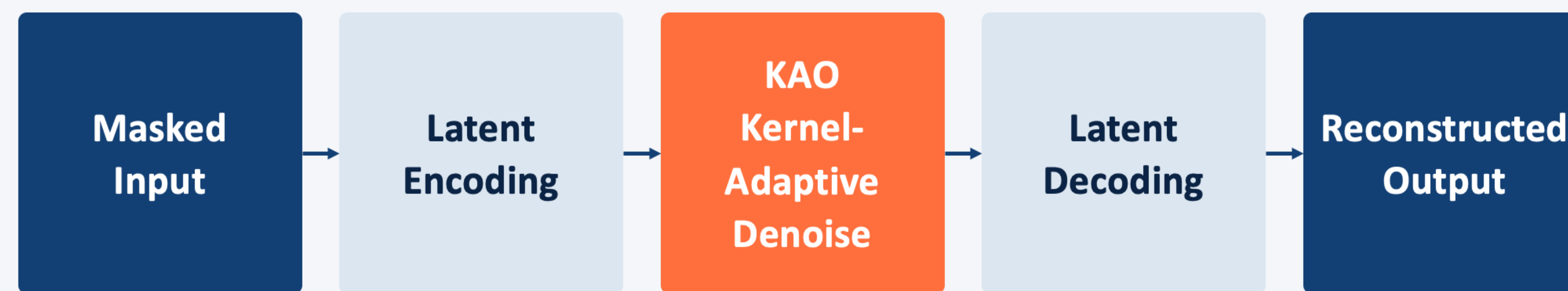
Teerapong Panboonyuen

<https://kaopanboonyuen.github.io/KAO>

Abstract

Satellite image inpainting is essential for restoring missing regions in remote sensing images. We propose **KAO**, a diffusion-based framework using **Kernel-Adaptive Optimization** and Latent Space Conditioning to achieve efficient and accurate VHR satellite image restoration. KAO improves inpainting performance through Explicit Propagation and forward-backward fusion on challenging datasets such as DeepGlobe and Massachusetts Roads Dataset.

Diffusion Models for Satellite Image Inpainting



GitHub

Qualitative Comparison Of Satellite Image Inpainting Across Seven Models



Satellite Mask

Stable Diffusion [16]

RePaint [12]

SatDiff [14]

DPS [3]

PSLD [17]

KAO (Ours)

KAO: Kernel-Adaptive Optimization for Satellite Image Inpainting

Algorithm 1 KAO: Kernel-Adaptive Optimization for Satellite Image Inpainting with Post-Conditioning via Token Pyramid Transformer (TPT).

Require: Diffusion model ($\mu_\theta(\cdot), \Sigma_\theta(\cdot)$), input satellite image x_0 , number of diffusion steps T , mask m , and set of latent tokens $\mathcal{H}$

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1:  $x_T \leftarrow$  sample from  $\mathcal{N}(0, I)$ 
2: for  $t$  from  $T$  to 1 do
3:   Step 1: Kernel-Adaptive Denoising (p-sample)
4:    $\mu^*, \Sigma^* \leftarrow \mu_\theta(x_t), \Sigma_\theta(x_t)$ 
5:    $x_{t-1}^{infr} \leftarrow$  sample from  $\mathcal{N}(\mu^*, \Sigma^*)$ 
6:   Step 2: Adaptive Noisy Condition Estimation (q-sample)
7:    $\mu, \Sigma \leftarrow \mu_q(x_0), \Sigma_q(x_0)$ 
8:    $x_{t-1}^{cond} \leftarrow$  sample from  $\mathcal{N}(\mu, \Sigma)$ 
9:   Step 3: Post-Conditioning via KAO in Latent Space
10:  for  $h$  in  $\mathcal{H}$  do
11:    Token-wise Adaptive Conditioning via TPT
12:     $h^* \leftarrow h^{infr} \odot (1 - D(m)) + h^{cond} \odot D(m)$ 
13:    Kernel-Guided Feature Propagation via TPT
14:     $\hat{h} \leftarrow \text{TPT}_\gamma^{-1}[\phi[\omega; \text{TPT}_\gamma(D(m), h^{cond})]]$ 
15:  end for
16:  Step 4: Kernel-Blended Reconstruction of Satellite Image
17:   $x_{t-1} \leftarrow x_{t-1}^{infr} \odot (1 - m) + x_{t-1}^{cond} \odot m$ 
18: end for
19: return  $x_0$   $\triangleright$  Final KAO-based satellite image reconstruction

```

Results

Massachusetts Roads Dataset			
Method	FID↓	Prec.↑	Rec.↑
Stable Diffusion	6.98	0.59	0.69
RePaint	6.12	0.65	0.71
LatentPaint	4.44	0.71	0.81
SatDiff	3.99	0.88	0.86
DPS	3.67	0.89	0.87
PSLD	3.42	0.91	0.89
KAO (Ours)	3.11	0.93	0.91

DeepGlobe 2018 Dataset			
Method	FID↓	Prec.↑	Rec.↑
Stable Diffusion	5.62	0.51	0.44
RePaint	5.19	0.59	0.47
LatentPaint	2.55	0.61	0.51
SatDiff	1.98	0.80	0.55
DPS	1.76	0.82	0.56
PSLD	1.65	0.84	0.58
KAO (Ours)	1.42	0.88	0.63

↓9%

FID VS. PSLD (MASS.)

↓14%

FID VS. PSLD (DEEPGLOBE)

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METRICS WON ON BOTH SETS



Reference

Panboonyuen, Teerapong. "KAO: Kernel-Adaptive Optimization in Diffusion for Satellite Image." **IEEE Transactions on Geoscience and Remote Sensing (IEEE TGRS 2025)**.

Contact: teerapong.panboonyuen@gmail.com, **Address:** Chulalongkorn University | Bangkok